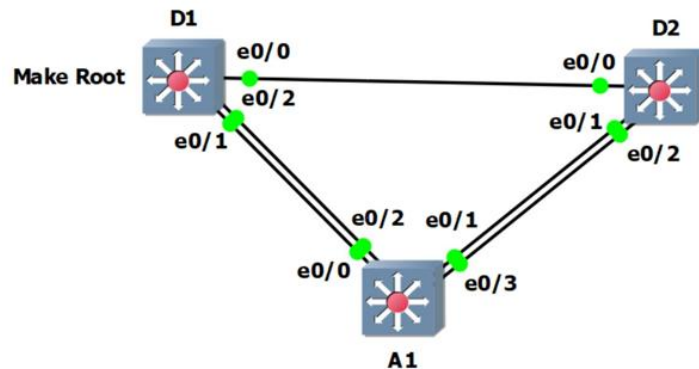


Practical No 5

Topology



Addressing Table

Device	Interface	IPv4 Address
D1	VLAN1	10.0.0.1/8
D2	VLAN1	10.0.0.2/8
A1	VLAN1	10.0.0.3/8

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Observe STP Convergence and Topology Change

Part 3: Configure and Verify Rapid Spanning Tree

Background / Scenario

The potential effect of a loop in the Layer 2 network is significant. Layer 2 loops could impact connected hosts as well as the network equipment. Layer 2 loops can be prevented by following good design practices and careful implementation of the Spanning Tree Protocol. In this lab, you will observe the operation of spanning tree protocols to protect the Layer 2 network from loops and topology disruptions. The terms "switch" and "bridge" will be used interchangeably throughout the lab.

Switch D1

enable

configure terminal

hostname D1

```
line console 0
logging synchronous
exec-timeout 0 0
spanning-tree mode pvst
interface vlan1
ip address 10.0.0.1 255.0.0.0
no shutdown
! ---- Uplinks ----
interface ethernet0/0
description to D2 e0/0
switchport
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
interface ethernet0/1
description to A1 e0/2
switchport
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
interface ethernet0/2
description to A1 e0/0
switchport
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
end
write memory
```

```

D1#en
D1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
D1(config)#hostname D1
D1(config)#line console 0
D1(config-line)#logging synchronous
D1(config-line)#exec-timeout 0 0
D1(config-line)#spanning-tree mode pvst
D1(config)#interface vlan1
D1(config-if)#ip address 10.0.0.1 255.0.0.0
D1(config-if)#no shutdown
D1(config-if)#
*Sep 25 14:21:01.151: %LINK-3-UPDOWN: Interface Vlan1, changed state to up
*Sep 25 14:21:02.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
D1(config-if)#interface ethernet0/0
D1(config-if)#description to D2 e0/0
D1(config-if)#switchport
D1(config-if)#switchport trunk encapsulation dot1q
D1(config-if)#switchport mode trunk
D1(config-if)#no shutdown
D1(config-if)#interface ethernet0/1
D1(config-if)#description to A1 e0/2
D1(config-if)#switchport
D1(config-if)#switchport trunk encapsulation dot1q
D1(config-if)#switchport mode trunk
D1(config-if)#no shutdown
D1(config-if)#interface ethernet0/2
D1(config-if)#description to A1 e0/0
D1(config-if)#switchport
D1(config-if)#switchport trunk encapsulation dot1q
D1(config-if)#switchport mode trunk
D1(config-if)#no shutdown
D1(config-if)#end
D1#
*Sep 25 14:23:03.083: %SYS-5-CONFIG_I: Configured from console by console
D1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
Compressed configuration from 1775 bytes to 1018 bytes[OK]
D1#

```

Make D1 a deterministic root

conf t

spanning-tree vlan 1 root primary

write memory

```

D1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
D1(config)#spanning-tree vlan 1 root primary
D1(config)#end
D1#wr
Building configuration...
Compressed configuration from 1813 bytes to 1041 bytes[OK]
D1#
*Sep 25 14:29:00.963: %SYS-5-CONFIG_I: Configured from console by console
D1#

```

Switch D2

enable

```
configure terminal
hostname D2
!
line console 0
logging synchronous
exec-timeout 0 0
!
spanning-tree mode pvst
!
Vlan 1
name management
interface vlan1
ip address 10.0.0.2 255.0.0.0
no shutdown
!
! ---- Uplinks ----
interface ethernet0/0
description to D1 e0/0
switchport
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
!
interface ethernet0/1
description to A1 e0/1
switchport
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
!
interface ethernet0/2
description to A1 e0/3
switchport
switchport trunk encapsulation dot1q
```

switchport mode trunk

no shutdown

!

end

write memory

```
D2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#hostname D2
D2(config)#line console 0
D2(config-line)#logging synchronous
D2(config-line)#exec-timeout 0 0
D2(config-line)#spanning-tree mode pvst
D2(config)#Vlan 1
D2(config-vlan)#name management
%Default VLAN 1 may not have its name changed.
D2(config-vlan)#interface vlan1
D2(config-if)#ip address 10.0.0.2 255.0.0.0
D2(config-if)#no shutdown
D2(config-if)#
*Sep 25 14:31:19.516: %LINK-3-UPDOWN: Interface Vlan1, changed state to up
*Sep 25 14:31:20.524: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
D2(config-if)#interface ethernet0/0
D2(config-if)#description to D1 e0/0
D2(config-if)#switchport
D2(config-if)#switchport trunk encapsulation dot1q
D2(config-if)#switchport mode trunk
D2(config-if)#no shutdown
D2(config-if)#interface ethernet0/1
D2(config-if)#description to A1 e0/1
D2(config-if)# switchport
D2(config-if)# switchport trunk encapsulation dot1q
D2(config-if)#switchport mode trunk
D2(config-if)#no shutdown
D2(config-if)#interface ethernet0/2
D2(config-if)#description to A1 e0/3
D2(config-if)#switchport
D2(config-if)#switchport trunk encapsulation dot1q
D2(config-if)#switchport mode trunk
D2(config-if)# no shutdown
D2(config-if)#no shutdown
D2(config-if)#end
```

Switch A1

enable

configure terminal

hostname A1

no ip domain-lookup

service password-encryption

banner motd # STP/RSTP Lab - A1 #

!

line console 0

logging synchronous

exec-timeout 0 0

```
!  
spanning-tree mode pvst  
!  
interface vlan1  
ip address 10.0.0.3 255.0.0.0  
no shutdown  
!  
! ---- Uplinks ----  
interface ethernet0/0  
description to D1 e0/2  
switchport  
switchport trunk encapsulation dot1q  
switchport mode trunk  
no shutdown  
!  
interface ethernet0/1  
description to D2 e0/1  
switchport  
switchport trunk encapsulation dot1q  
switchport mode trunk  
no shutdown  
!  
interface ethernet0/2  
description to D1 e0/1  
switchport  
switchport trunk encapsulation dot1q  
switchport mode trunk  
no shutdown  
!  
interface ethernet0/3  
description to D2 e0/2  
switchport  
switchport trunk encapsulation dot1q  
switchport mode trunk
```

no shutdown

!

end

write memory

```
A1#enable
A1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
A1(config)#hostname A1
A1(config)#no ip domain-lookup
A1(config)#service password-encryption
A1(config)#banner motd # STP/RSTP Lab - A1 #
A1(config)#line console 0
A1(config-line)#logging synchronous
A1(config-line)#exec-timeout 0 0
A1(config-line)#spanning-tree mode pvst
A1(config)#interface vlan1
A1(config-if)#ip address 10.0.0.3 255.0.0.0
A1(config-if)#no shutdown
A1(config-if)#
*Sep 25 14:37:41.377: %LINK-3-UPDOWN: Interface Vlan1, changed state to up
*Sep 25 14:37:42.378: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
A1(config-if)#interface ethernet0/0
A1(config-if)#description to D1 e0/2
A1(config-if)#switchport
A1(config-if)#switchport trunk encapsulation dot1q
A1(config-if)#switchport mode trunk
A1(config-if)#no shutdown
A1(config-if)#interface ethernet0/1
A1(config-if)#description to D2 e0/1
A1(config-if)#switchport
A1(config-if)#switchport trunk encapsulation dot1q
A1(config-if)#switchport mode trunk
A1(config-if)#no shutdown
A1(config-if)#interface ethernet0/2
A1(config-if)#description to D1 e0/1
A1(config-if)#switchport
A1(config-if)#switchport trunk encapsulation dot1q
A1(config-if)#switchport mode trunk
A1(config-if)#no shutdown
A1(config-if)#interface ethernet0/3
A1(config-if)#description to D2 e0/2
A1(config-if)#switchport
A1(config-if)#switchport trunk encapsulation dot1q
A1(config-if)#switchport mode trunk
A1(config-if)#no shutdown
A1(config-if)#end
A1#
*Sep 25 14:40:00.300: %SYS-5-CONFIG_I: Configured from console by console
A1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
Compressed configuration from 1891 bytes to 1085 bytes[OK]
```

4) Sanity Checks (what/why/how)

D1#show interfaces trunk ! verify trunks are up and VLAN 1 is allowed/active

```
D1#show interfaces trunk ! verify trunks are up and VLAN 1 is allowed/active

Port      Mode      Encapsulation  Status        Native vlan
Et0/0     on        802.1q         trunking      1
Et0/1     on        802.1q         trunking      1
Et0/2     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Et0/0     1-4094
Et0/1     1-4094
Et0/2     1-4094

Port      Vlans allowed and active in management domain
Et0/0     1
Et0/1     1
Et0/2     1

Port      Vlans in spanning tree forwarding state and not pruned
Et0/0     1
Et0/1     1
Et0/2     1
D1#
```

D2#show interfaces trunk

```
D2#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
Et0/0     on        802.1q         trunking      1
Et0/1     on        802.1q         trunking      1
Et0/2     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Et0/0     1-4094
Et0/1     1-4094
Et0/2     1-4094

Port      Vlans allowed and active in management domain
Et0/0     1
Et0/1     1
Et0/2     1

Port      Vlans in spanning tree forwarding state and not pruned
Et0/0     1
Et0/1     1
Et0/2     1
```

A1#show interfaces trunk

```
A1#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
Et0/0     on        802.1q         trunking      1
Et0/1     on        802.1q         trunking      1
Et0/2     on        802.1q         trunking      1
Et0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Et0/0     1-4094
Et0/1     1-4094
Et0/2     1-4094
Et0/3     1-4094

Port      Vlans allowed and active in management domain
Et0/0     1
Et0/1     1
Et0/2     1
Et0/3     1

Port      Vlans in spanning tree forwarding state and not pruned
Et0/0     1
Et0/1     none
Et0/2     none
Et0/3     none
```

D1#show spanning-tree summary

```
D1#show spanning-tree summary
Switch is in pvst mode
Root bridge for: VLAN0001
EtherChannel misconfig guard is enabled
Extended system ID      is enabled
Portfast Default        is disabled
Portfast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

Name                    Blocking Listening Learning Forwarding STP Active
-----
VLAN0001                0         0         0         16         16
-----
1 vlan                  0         0         0         16         16
```

D2#show spanning-tree summary


```
D2#show spanning-tree summary
Switch is in pvst mode
Root bridge for: none
EtherChannel misconfig guard is enabled
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	16	16
1 vlan	0	0	0	16	16

A1#show spanning-tree summary

```
A1#show spanning-tree summary
Switch is in pvst mode
Root bridge for: none
EtherChannel misconfig guard is enabled
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	3	0	0	13	16
1 vlan	3	0	0	13	16

(From D1 for example PING):

D1#ping 10.0.0.2 source vlan1 ! reach D2 SVI

```
D1#ping 10.0.0.2 source vlan1 ! reach D2 SVI
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
Packet sent with a source address of 10.0.0.1
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
```

5) Part 2 -Discover the Default Spanning Tree (PVST+) (Per-VLAN Spanning Tree Plus)

A) Identify the Root Bridge

D1#show spanning-tree root

```
D1#show spanning-tree root
```

Vlan	Root ID	Root Cost	Hello Time	Max Age	Fwd Dly	Root Port
VLAN0001	24577 aabb.cc00.0100	0	2	20	15	

Record: Bridge ID of the root, Root Port (local), and Root Path Cost. If you set root primary on

D1 should be the root.

B) Inspect Port Roles/States

- **Root Bridge (likely D1):** all participating ports should be Designated Forwarding.
- **Non-root (A1, D2):** each has exactly one Root port; redundant paths become Alternate (Blocking/Discarding). Which one blocks depends on root path cost and tie-breakers (bridge/port IDs).

D1#show spanning-tree vlan 1

```
D1#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
            Address     aabb.cc00.0100
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
            Address     aabb.cc00.0100
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Desg FWD 100      128.1   Shr
Et0/1                    Desg FWD 100      128.2   Shr
Et0/2                    Desg FWD 100      128.3   Shr
Et0/3                    Desg FWD 100      128.4   Shr
Et1/0                    Desg FWD 100      128.5   Shr
Et1/1                    Desg FWD 100      128.6   Shr
Et1/2                    Desg FWD 100      128.7   Shr
Et1/3                    Desg FWD 100      128.8   Shr
Et2/0                    Desg FWD 100      128.9   Shr
Et2/1                    Desg FWD 100      128.10  Shr
Et2/2                    Desg FWD 100      128.11  Shr
Et2/3                    Desg FWD 100      128.12  Shr
Et3/0                    Desg FWD 100      128.13  Shr
Et3/1                    Desg FWD 100      128.14  Shr
Et3/2                    Desg FWD 100      128.15  Shr
Et3/3                    Desg FWD 100      128.16  Shr
```

D2#show spanning-tree vlan 1

```
D2#show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    24577
           Address    aabb.cc00.0100
           Cost      100
           Port      1 (Ethernet0/0)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    aabb.cc00.0200
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Root	FWD	100	128.1	Shr
Et0/1	Desg	FWD	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr
Et0/3	Desg	FWD	100	128.4	Shr
Et1/0	Desg	FWD	100	128.5	Shr
Et1/1	Desg	FWD	100	128.6	Shr
Et1/2	Desg	FWD	100	128.7	Shr
Et1/3	Desg	FWD	100	128.8	Shr
Et2/0	Desg	FWD	100	128.9	Shr
Et2/1	Desg	FWD	100	128.10	Shr
Et2/2	Desg	FWD	100	128.11	Shr
Et2/3	Desg	FWD	100	128.12	Shr
Et3/0	Desg	FWD	100	128.13	Shr
Et3/1	Desg	FWD	100	128.14	Shr
Et3/2	Desg	FWD	100	128.15	Shr
Et3/3	Desg	FWD	100	128.16	Shr

```
A1#show spanning-tree vlan 1
```

```
A1#show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    24577
           Address    aabb.cc00.0100
           Cost      100
           Port      1 (Ethernet0/0)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    aabb.cc00.0300
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Root	FWD	100	128.1	Shr
Et0/1	Altn	BLK	100	128.2	Shr
Et0/2	Altn	BLK	100	128.3	Shr
Et0/3	Altn	BLK	100	128.4	Shr
Et1/0	Desg	FWD	100	128.5	Shr
Et1/1	Desg	FWD	100	128.6	Shr
Et1/2	Desg	FWD	100	128.7	Shr
Et1/3	Desg	FWD	100	128.8	Shr
Et2/0	Desg	FWD	100	128.9	Shr
Et2/1	Desg	FWD	100	128.10	Shr
Et2/2	Desg	FWD	100	128.11	Shr
Et2/3	Desg	FWD	100	128.12	Shr
Et3/0	Desg	FWD	100	128.13	Shr
Et3/1	Desg	FWD	100	128.14	Shr
Et3/2	Desg	FWD	100	128.15	Shr
Et3/3	Desg	FWD	100	128.16	Shr

6) Show It Adjusts -Simulate a Failure (PVST+)

Turn on event logs (D2) and

Fail a link (shut D2 A1 on D2 e0/1):

```
debug spanning-tree events
```

```
conf t
```

```
interface ethernet0/1
```

shutdown

end

```
D2#debug spanning-tree events
Spanning Tree event debugging is on
D2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#interface ethernet0/1
D2(config-if)#shutdown
D2(config-if)#en
*Sep 25 14:53:43.796: STP: VLAN0001 sent Topology Change Notice on Et0/0
*Sep 25 14:53:43.796: STP[1]: Generating TC trap for port Ethernet0/1
D2(config-if)#end
D2#
*Sep 25 14:53:45.796: %LINK-5-CHANGED: Interface Ethernet0/1, changed state to administratively down
*Sep 25 14:53:46.799: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to down
D2#
*Sep 25 14:53:47.255: %SYS-5-CONFIG_I: Configured from console by console
D2#
```

Observe:

- STP re-computes; a previously Alternate link should unblock to maintain a single loop-free path.
- PVST+ takes several seconds (listening/learning) before forwarding resumes.

Restore link & stop logs:

D2:

conf t

interface ethernet0/1

no shutdown

end

undebug all

```
D2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#interface ethernet0/1
D2(config-if)#no shutdown
D2(config-if)#
*Sep 25 14:55:07.755: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Sep 25 14:55:08.761: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
*Sep 25 14:55:08.763: set portid: VLAN0001 Et0/1: new port id 8002
*Sep 25 14:55:08.763: STP: VLAN0001 Et0/1 -> listening
D2(config-if)#end
D2#
*Sep 25 14:55:12.610: %SYS-5-CONFIG_I: Configured from console by console
D2#undebug all
All possible debugging has been turned off
D2#
```

Re-check roles:

D2#show spanning-tree vlan 1

```
D2#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     aabb.cc00.0100
             Cost       100
             Port       1 (Ethernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.0200
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Root FWD 100       128.1   Shr
Et0/1                    Desg FWD 100       128.2   Shr
Et0/2                    Desg FWD 100       128.3   Shr
Et0/3                    Desg FWD 100       128.4   Shr
Et1/0                    Desg FWD 100       128.5   Shr
Et1/1                    Desg FWD 100       128.6   Shr
Et1/2                    Desg FWD 100       128.7   Shr
Et1/3                    Desg FWD 100       128.8   Shr
Et2/0                    Desg FWD 100       128.9   Shr
Et2/1                    Desg FWD 100       128.10  Shr
Et2/2                    Desg FWD 100       128.11  Shr
Et2/3                    Desg FWD 100       128.12  Shr
Et3/0                    Desg FWD 100       128.13  Shr
Et3/1                    Desg FWD 100       128.14  Shr
Et3/2                    Desg FWD 100       128.15  Shr
Et3/3                    Desg FWD 100       128.16  Shr
```

7) Migrate to RSTP (Rapid-PVST) and Compare

Enable on all switches:

D1:

conf t

spanning-tree mode rapid-pvst

end

write memory

```
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#spanning-tree mode rapid-pvst
D1(config)#end
D1#
*Sep 25 14:57:00.593: %SYS-5-CONFIG_I: Configured from console by console
D1#
```

D1#show spanning-tree summary ! should indicate rapid-pvst

```
D1#show spanning-tree summary ! should indicate rapid-pvst
Switch is in rapid-pvst mode
Root bridge for: VLAN0001
EtherChannel misconfig guard is enabled
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

Name                Blocking Listening Learning Forwarding STP Active
-----
VLAN0001              0          0          0          16          16
1 vlan                0          0          0          16          16
```

D2:

```
conf t
spanning-tree mode rapid-pvst
end
write memory
```

```
D2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#spanning-tree mode rapid-pvst
D2(config)#end
D2#
```

D2#show spanning-tree summary ! should indicate rapid-pvst

```
D2#show spanning-tree summary ! should indicate rapid-pvst
Switch is in rapid-pvst mode
Root bridge for: none
EtherChannel misconfig guard is enabled
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

Name Blocking Listening Learning Forwarding STP Active
-----
VLAN0001 0 0 0 16 16
-----
1 vlan 0 0 0 16 16
```

A1#configure terminal

A1(config)#spanning-tree mode rapid-pvst

A1(config)#end

```
A1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
A1(config)#spanning-tree mode rapid-pvst
A1(config)#end
A1#
*Sep 25 15:00:20.513: %SYS-5-CONFIG_I: Configured from console by console
A1#
```

A1#show spanning-tree summary

```
A1#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: none
EtherChannel misconfig guard is enabled
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

Name Blocking Listening Learning Forwarding STP Active
-----
VLAN0001 3 0 0 13 16
-----
1 vlan 3 0 0 13 16
A1#
```

Repeat the same failure test (again D2 e0/1):

D2#debug spanning-tree events

D2#configure terminal

D2(config)#interface e0/1

D2(config-if)#shutdown

D2(config-if)#end

```
D2#debug spanning-tree events
Spanning Tree event debugging is on
D2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#interface e0/1
D2(config-if)#shutdown
D2(config-if)#end
D2#
*Sep 25 15:00:31.231: %SYS-5-CONFIG_I: Configured from console by console
*Sep 25 15:00:31.478: %LINK-5-CHANGED: Interface Ethernet0/1, changed state to administratively down
*Sep 25 15:00:32.484: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to down
D2#
*Sep 25 15:03:41.061: RSTP(1): Et0/0 rcvd info expired
*Sep 25 15:03:41.061: RSTP(1): updt roles, information on root port Et0/0 expired
*Sep 25 15:03:41.061: RSTP(1): we become the root bridge
*Sep 25 15:03:41.061: RSTP(1): Et0/0 is now designated
*Sep 25 15:03:41.076: RSTP(1): updt roles, received superior bpdu on Et0/0
*Sep 25 15:03:41.076: RSTP(1): Et0/0 is now root port
*Sep 25 15:03:41.076: STP[1]: Generating TC trap for port Ethernet0/0
```

Expected: Noticeably faster reconvergence (often sub-second to a couple of seconds on point-to-point links), thanks to RSTP's proposal/agreement and sync processes.

D2#configure terminal

D2(config)#interface e0/1

D2(config-if)#no shutdown

D2(config-if)#end

D2#undebug all

```
D2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
D2(config)#interface e0/1
D2(config-if)#no shutdown
D2(config-if)#
*Sep 25 15:06:59.994: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Sep 25 15:07:00.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
*Sep 25 15:07:00.996: RSTP(1): initializing port Et0/1
*Sep 25 15:07:00.996: RSTP(1): Et0/1 is now designated
*Sep 25 15:07:00.999: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:02.951: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:04.961: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:06.966: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:08.977: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:10.986: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
*Sep 25 15:07:13.000: RSTP(1): transmitting a proposal on Et0/1
D2(config-if)#end
```

Results Checked

PVST+ took approx. 16 microseconds

RSTP took approx. 7 microseconds